Viransh Shah

B.A.Sc. Computer Science — McMaster University 2026

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Education

McMaster University Sept 2021 – June 2026

Bachelor of Applied Science in Computer Science, Minor in Statistics *Hamilton, Ontario*

- Dean's Honours List, GPA 3.7 / 4.0 in Information Security, Computer Networks & Security, Databases, and Algorithms.

Experience

McMaster Cyber Society Sept 2023 – Present

Founding Executive Member *Hamilton, Ontario*

- Placed **3rd** as MacHackers at **CyberSci Regional Finals 2025** and **2nd** in Toronto against top Canadian university teams (photo); also competed in Advent of Cyber and CyberStart Canada.
- Reconstructed attack chains from PCAP captures, memory images, and file metadata using Wireshark, tcpdump, and CyberChef.
- Cracked weak and reused password hashes with John the Ripper and reproduced brute-force and password-spraying traffic with Hydra to study its detection signature; exploited SQL injection and authentication bypass.

Environment and Climate Change Canada & Canada Water Agency Sept 2024 – Aug 2025

Full Stack Developer – IMWEBs Viewer & Nutri View *Burlington & North York, Ontario*

- Sole developer of IMWEBs Viewer and Nutri View from prototype to production: a cross-platform desktop and web application (Vue.js, Tauri, Flask) giving federal scientists self-serve access to watershed data.
- Secured access with bcrypt-hashed credentials, session tokens with server-side revocation, and role-based access control defaulting to deny; hardened input against SQL injection and path traversal.
- Automated Excel and water quality data conversion into SQLite, cutting analyst data preparation from hours to zero; automated signed cross-platform builds and deployed over HTTPS behind IIS on Azure VMs (ECCC-IMWEBs-Viewer, Nutri-View).

McMaster University Jan–Apr 2024; Sep–Dec 2025; Jan–Apr 2026

Teaching Assistant – SFWRENG 2C03, 2XC3 & COMPSCI 1XD3 *Hamilton, Ontario*

- Led labs and tutorials for 100+ students across three terms on algorithms, data structures, and software development.

Security Training (TryHackMe)

Pre Security and Cyber Security 101 paths – 65+ rooms 2025

Certificates: Pre Security, Cyber Security 101

- Investigated alerts end to end in **Splunk**, pivoting from a suspicious process to its hash, parent process, and scheduled-task persistence on a Windows host.
- Triaged **Windows Security event logs** for logon activity, process creation, account and group changes, and service installation, and Linux `/var/log/auth.log` SSH authentication trails.
- Analyzed network and email protocols with Wireshark, tcpdump, and Nmap; read `Received` chains and SPF, DKIM, and DMARC results for phishing triage.

Projects

Slicc | *Bill Splitting and Restaurant Recommendation App* | GitHub Sept 2025 – Apr 2026

- Cross-platform mobile app (React Native, Expo, TypeScript, Firebase) built by a 7-person Agile capstone team and released to Android.
- Architected **libsodium/NaCl end-to-end encryption** with device-held private keys, so messages are never readable by the server or Firebase backend.

Technical Skills

Security Operations: Alert triage, incident response (NIST lifecycle), log and packet analysis, Windows Security event logs, Linux syslog, SIEM and EDR investigation, email header analysis, Active Directory, TCP/IP
Security Tools: Splunk, Wireshark, tcpdump, CyberChef, Event Viewer, Nmap; from CTF practice: Burp Suite, Metasploit, John the Ripper, Hydra, SQLMap, Kali Linux
Databases & Platforms: Microsoft SQL Server, PostgreSQL, MySQL, SQLite, Azure, IIS, GNU/Linux, Windows, Docker, Git
Languages: Python, PowerShell, Bash, SQL, TypeScript, JavaScript, Java, C